

DEIDINE CHEIGEUR

Software Engineer

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PROFESSIONAL SUMMARY

Software engineer with experience delivering full-stack web, mobile, and desktop applications across real estate, e-commerce, government, and fintech-adjacent projects. Skilled in React/Next.js, React Native, Spring Boot, and Python, with hands-on experience in AI integration (OpenAI SDK, RAG), Kubernetes-based cloud deployment (AWS, GCP, Docker, Helm), OpenID Connect/SSO identity systems, and both SQL and NoSQL database design.

SKILLS

Languages: JavaScript, TypeScript, Python, Java, C, C++, PHP, Dart, SQL

Frontend & Mobile: React, Next.js, Angular, Vue.js, React Native, Electron, HTML, CSS, Tailwind CSS, Bootstrap

Backend: Spring Boot, Hibernate/JPA, Laravel, Django, REST API, SOAP

Databases: MySQL, PostgreSQL, MongoDB, Firebase, Supabase, SQL Server, NoSQL

AI & Data: Machine Learning, Deep Learning, NLP, OpenAI SDK, Retrieval-Augmented Generation (RAG), Image Processing

Real-Time & Native: WebRTC, LiveKit, Socket.IO, end-to-end encryption, native React Native/Expo modules, Turborepo/Yarn Berry monorepos

Identity & SSO: OpenID Connect (OIDC), OAuth2, PKCE, Koa, oidc-provider, multi-service identity brokering

Cloud & DevOps: AWS, Google Cloud, Docker, Kubernetes, Helm, Nginx, Caddy, CI/CD, Git

Security & Networking: Wireshark, Burp Suite, Nmap, Kali Linux, network fundamentals

Tools & Methods: Figma, Postman, QGIS, Odoo, Claude (AI-assisted development), UML, Agile, Merise (MCD/MLD/MCT), Design Patterns

Release & Deployment: Google Play Console, Apple App Store Connect / TestFlight, CodePush OTA updates, Docker, GitHub Actions CI/CD, Fastlane-style versioned release pipelines

Digital Marketing: Facebook & LinkedIn Page management, Meta Ads Manager campaign creation, social media marketing

PROJECT EXPERIENCE (FREELANCE)

Nougta – Multi-Tenant Point-of-Sale System

2025 – Present

- Building an offline-first POS mobile app (React Native, WatermelonDB) with real-time multi-device sync via Firebase Firestore and FCM.
- Developed the NestJS/MongoDB backend and Next.js admin dashboard, ecommerce storefront, and accounting module.
- Own end-to-end release management: versioned AAB/APK and iOS release builds, Play Store and App Store Connect submissions, CodePush OTA updates, and a custom version-guard script enforcing Play Store/App Store versioning rules in CI.
- Set up Docker-based deployment and GitHub Actions CI/CD pipelines for the backend and web apps.

Enterprise Communication Platform (Confidential) – Native E2E Encryption & WebRTC Modules

2025 – Present

- Contributed native React Native/Expo modules to a whitelabel, multi-client secure communication platform (institutional clients), focused on end-to-end encrypted peer-to-peer and group voice/video calling and messaging.
- Built real-time WebRTC calling features on top of LiveKit and Socket.IO signaling for P2P and group calls.
- Built a daily reporting feature letting each user submit a full activity report, aggregated for operational review.
- Worked within a Yarn/Turborepo monorepo alongside dedicated mobile, dashboard, and infrastructure teams.

Khidmaty – National Government Services Platform (Mobile, Backend & Ops)

Nov 2024 – Present

- Own end-to-end maintenance of a national government services platform serving 500,000+ users, covering vehicle registration/ownership transfer (ANRPTS, carte grise) and judicial record (casier judiciaire) requests with in-app PDF preview, share, and download.
- Built and operate a custom OpenID Connect (OIDC) identity provider (Koa, oidc-provider, PKCE) so Khidmaty serves as the shared identity provider (SSO) for other Mauritanian government digital services.
- Led the full backend migration of Mauritania's Carte Grise (vehicle registration) system onto a Next.js API architecture with a complete certificate state-machine workflow, integrating it with both the Khidmaty and Houwity apps.
- Deploy and operate Kubernetes clusters (Docker, Helm charts) across multiple government projects, configuring SSL/HTTPS via Nginx and Caddy.
- Administer the production database, including regular backups and data integrity checks.
- Release and maintain the app on both Google Play and the Apple App Store; resolve store policy reviews and compliance issues to keep releases approved and live.
- Triage and fix client-reported bugs and mobile app crashes/memory issues; monitor the platform for security threats and apply patches; upgrade dependencies and OS/SDK targets to the latest supported versions.
- Represent the engineering side in meetings with ministry officials and senior government stakeholders on requirements, incidents, and roadmap.

Houwity – National Registry Mobile App (Frontend)

Nov 2024 – Present

- Built and maintain the mobile app frontend for Houwity, a national registry platform equivalent to Mauritania's official ANRPTS portal (anrpts.gov.mr), giving citizens mobile access to the same registry services as the Khidmaty app.
- Connected Houwity to the migrated Carte Grise backend and the shared Khidmaty OIDC identity provider for unified authentication across registry services.
- Own ongoing maintenance: feature updates, bug fixes, and alignment with backend/API changes as the registry system evolves.

Temwine – Grocery Supply Chain Mobile App

Oct 2024 – Present

- Developing a React Native app with a Next.js/MongoDB backend to help low-income families purchase government-subsidized groceries.
- Built APIs to verify beneficiary eligibility against the national ID registry and enforce one-purchase-per-day limits.
- Implemented OTP verification and shipment notifications for store owners.

Real Estate Listing Platform

2024

- Built a property listing website with city-based filtering for rentals and sales.

- Improved SEO and set up analytics tracking (TTFB, FID, FCP) with Matomo and Google Analytics to monitor user behavior and bounce rate.
- Technologies: Next.js, Supabase, Tailwind CSS.

AI Chatbot with Retrieval-Augmented Generation 2024

- Built a GPT-based chat application with real-time response streaming using the OpenAI SDK.
- Implemented RAG to answer company-specific and general-knowledge queries.
- Technologies: Next.js, Supabase.

E-Commerce Platform Mar – Apr 2024

- Built an e-commerce site with product browsing, cart, and checkout, plus an admin dashboard for product, inventory, and warehouse management.
- Deployed to AWS and Hostinger and configured a VPS with a custom domain for production hosting.
- Technologies: Spring Boot, Hibernate/JPA, React.

Custom Form Builder 2023 – 2024

- Built a drag-and-drop form builder (file uploads, dates, checkboxes, radio buttons, text fields) with live style customization.
- Added code export to Flutter, React, Next.js, and HTML, plus form sharing and submission collection.
- Technologies: Next.js, Supabase.

Stadium Booking Mobile App Sep – Dec 2023

- Built a mobile app for booking football stadiums, with availability browsing and payment-proof upload for owner validation.
- Built owner-side tools for managing stadiums and tracking reservations.
- Technologies: Flutter, Dart, Django, MySQL, REST API, GetX, Google Maps.

Point-of-Sale & Inventory Management System Jul 2023

- Built a desktop POS application supporting cash, credit, VAT, and bank payment types, with low-stock alerts and sales/profit reporting.
- Implemented client debt tracking, transaction history, and automated database backups.
- Technologies: Java, Swing, MySQL, Hibernate, MVC.

Bus Fleet & Passenger Management System Jul 2023

- Built a desktop app for a bus agency to manage inter-city trips, seat booking, and worker/package tracking.
- Prevented duplicate seat assignments and generated printable passenger manifests per trip.
- Technologies: Python, Tkinter, MySQL.

Freelance Marketplace Platform 2023

- Built a freelance marketplace connecting freelancers with clients, including profiles, job requests, and fee collection.
- Technologies: HTML, CSS, JavaScript, PHP, MySQL, Bootstrap.

Inventory Management System – Ministry of Technology Transformation 2023

- Developed an inventory tracking system for stock levels, transactions, and reporting for a government ministry.
- Technologies: Spring Boot, Hibernate/JPA, Angular.

Bus Travel Booking Website 2020

- Built a website for bus seat booking and passenger management, with printable passenger lists for security checkpoints.
- Hardened the application against SQL injection, XSS, and HTML injection.
- Technologies: Spring Boot, Hibernate/JPA, React, MySQL.

EDUCATION

Ph.D. in Secure Distributed Systems (2nd year) – University of Nouakchott	<i>2024 – Present</i>
Master's in Information Technology – University of Nouakchott	<i>2022 – 2024</i>
Bachelor's in Development Administration (Internet & Intranet) – University of Nouakchott	<i>2019 – 2022</i>
Baccalauréat, Mathematics (Bac C) – Lycée Atar	<i>2016 – 2019</i>

PROFESSIONAL TRAINING

- Artificial Intelligence, Office Automation, Business Management, and Soft Skills (2020 – 2023)
- Digital Marketing: Facebook & LinkedIn Page management and paid ad campaign creation (2020 – 2023)

LANGUAGES

Arabic (Native) • French (Fluent) • English (Fluent)